HARMANPREET KAUR

Ai & Machine Learning Undergraduate | Data Analytics | Generative AI | Frontend developer intern

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EDUCATION

- **Chandigarh University** |Mohali, Punjab — July 2024 -July2028
  B.E. in Computer Science and Engineering — **CGPA**: 8.35
- **Bhai Nand Lal Public School, Sri Anandpur Sahib** -2024
  CBSE– Class XII — **Percentage**: 81.7%
- **Shri Guru Ram Rai Public School, Bharatgarh** -2022
  CBSE– Class X — **Percentage**: 86.2%

EXPERIENCE

Frontend Developer Intern **/ *Trigunasoft (Remote)*** — July 2025 -July 2026

Tech: Angular, TypeScript, SignalR, JWT Auth, Angular Signals

- Developed **31+ components and 31 routed screens** across user, content, and admin modules.
- Implemented **JWT authentication and route guards**, securing 90%+ of application routes.
- Built **multilingual support, real-time chat using SignalR, and admin dashboards**.

PROJECTS

LOST & FOUND AI PLATFORM — Github

Tech: Next.js, FastAPI, MongoDB, WebSockets, Gemini API, Hugging Face

- Built an **AI-powered lost & found platform** using semantic, location, and image-based matching.
- Reduced **memory usage by 90%+** and improved **API performance by 40%** using Hugging Face/Gemini APIs, lazy loading, asynchronous FastAPI, and MongoDB Motor.
- Implemented **WebSocket-based real-time messaging** and secure claim verification, reducing messaging overhead by **85%** and spam by **~95%**.

Orbit AI | AI-Powered Multi-Agent Platform — Github

Tech*:* React, Node.js, LangGraph, LangChain, Qdrant, Redis, Docker, AWS

- Built 7+ AI workflows for research, coding, document generation, PDF RAG, and image processing using LangGraph and multiple LLM APIs.
- Implemented end-to-end RAG with recursive chunking, embeddings, and Qdrant vector search for context-grounded PDF question answering.
- Developed 5 microservices with secure authentication, Redis sessions, and CI/CD deployment using GitHub Actions, Docker, and AWS ECS/ECR.

Image Caption Generator — Github

Tech: Python, TensorFlow, CNN, LSTM

- Developed an image captioning system using CNN-LSTM to generate descriptive captions from images.
- Trained on the Flickr8K dataset using transfer learning, feature extraction, and sequence modeling.
- Built a web interface for real-time image upload and caption generation.

TECHNICAL & INTERPERSONAL SKILLS

- **Languages:** Python, C++, JavaScript, TypeScript, SQL
- **Web Development:** React, Next.js, Angular, HTML, CSS, Tailwind CSS, Bootstrap, Redux Toolkit,FastAPI, Node.js, Express.js, REST APIs, JWT Authentication, OAuth 2.0, WebSockets
- **AI/ML:** Data Preprocessing, EDA, Feature Engineering, Machine Learning, Deep Learning, NLP, Recommendation Systems, Model Evaluation, LLMs, Prompt Engineering, RAG, Agentic AI, LangChain, LangGraph, Embeddings, Vector Databases, Semantic Search, LLM APIs
- **Tools & Cloud:** Git, GitHub, Postman, VS Code, Docker, GitHub Actions, AWS (ECR, ECS, S3, CloudFront), CI/CD
- **Databases:** MongoDB, Microsoft SQL Server, ChromaDB, Qdrant, Redis
- **Interpersonal:** Communication, Team Collaboration, Leadership, Problem-Solving, Adaptability

ACHIEVEMENTS

- **Awarded a 100% merit-based scholarship** covering the entire four-year undergraduate program at Chandigarh University.
- **Top 5 AI/ML Hackathon Team** — Developed and presented an applied machine-learning solution in a competitive, team-based technical challenge.
- **Selected 1 of 121 from 500+ applicants** for the **Liaison Officer program at Chandigarh University**, representing the university in professional and corporate engagements.